

Some interesting wireless applications suggested by Electronics Talk

WigWag. WigWag lets you build intelligent environments by using Internet-enabled wireless sensors and devices. It uses very simple rules like, "When (this) happens, then do (that)." It enables home automation in a very simple way.

Twine. Durable 16.12 sq cm (2.5" square) internal and external sensors with Wi-Fi connectivity allow you to use wireless connected objects anywhere that Wi-Fi is available. It will alert you when an event happens; for instance, when the laundry gets done, the pet opens the front door, or anything else you can think of.

SmartThings. This is one of the easiest ways to connect real-world objects to the virtual world using wireless technology and the Internet.

Valta. The name gives a clue to what it does. This product is a wireless energy management system that automatically detects unused devices, identifies the energy wasted and notifies you with solutions.

Electronics Talk is a discussion board promoted by EFW. You can access it at www.forum.electronicsforu.com

Alternatives.

1. *Looking for lower implementation costs?* Classic Bluetooth is a good alternative for Wireless USB because of the extremely low cost of implementation, if you don't mind the lower speeds, that is.

2. *Need more recent technology?* The latest version of Wi-Fi (IEEE 802.11ac) is a good alternative, although it is yet to be ratified by the international body responsible – the IEEE.

Near Field Communication (NFC)

Designed as a personal area network (PAN) technology that allows devices such as smartphones to communicate over a small area, NFC has made it big in recent years by getting featured on a multitude of smartphones. It can also be used as a tag, where a powered NFC device can talk to an unpowered NFC tag and gather information from it. This opens the way to a number of innovative use cases which are not possible on other technologies.

NFC works on ISO/IEC 14443, ISO 18000-3, and those defined by the NFC Forum. Working in the 13.56MHz band, it has a range of less than 20 cm and a bit rate of 424 kbit/s.

The best fit. Contactless transactions, data exchange and short range authentication are some of the applications that best suit NFC. It can also be used as a simplified set-up of more complex communication technologies such as Bluetooth and Wi-Fi. NFC tags

can also be placed on posters, and programmed such that if a device touches them, they give you additional information such as maps, Web addresses and movie trailers, depending on the kind of poster or billboard.

Classic Bluetooth

Classic Bluetooth technology (v2.1 + EDR) is well suited for wireless integration of automation devices, but perhaps it is more famous for being the most used communication technology for transferring photos, videos, songs and other media over smartphones and tablets. It allows a gross data rate of up to 2.1 Mbit/s with a latency of around 10 ms.

Bluetooth is limited to lower data rates (typically 1 Mbit/s), which is enough for the transmission of voice and music data, for example. "Because WLAN can support rates of up to around 100 Mbit/s, it is suitable for downloading (or streaming) multimedia content such as movies. A typical example of using Bluetooth, cellular (GSM or UMTS), and WLAN simultaneously is taking a cellular call on a smartphone with a Bluetooth headset, while streaming a video on the Internet," explains Dr Patric Heide, head of the connectivity business division of the SAW Business Group at EPCOS AG, a group company of TDK Corporation.

The best fit. It is best suited for environments where multiple wireless devices can be connected in the

same radio environment and can still operate flawlessly. It is specified for devices that have to maintain a small footprint, low power consumption and be cost-efficient.

Alternative.

1. *Need a higher data rate?* You can check out Wi-Fi if you don't mind the higher power requirements.

Wi-Fi

This is the most common Wireless LAN technology. It was designed as a wireless extension to Ethernet, and is used for all sorts of applications that can use a high bandwidth and a longer range with a slight power penalty. It also has a built-in roaming function that reduces hand-off delays. It has a range of 200 metres in the 2.4GHz range and a data throughput of 11-54 Mbit/s for IEEE 802.11b/g. The latest, yet currently unrati-fied, version is IEEE 802.11ac. It isn't due for final ratification until 2014.

"5G Wi-Fi, the 5th generation of Wi-Fi based on the IEEE 802.11ac standard, delivers the world's fastest and most reliable wireless coverage for HD-quality video streaming and near instantaneous data synchronisation. Integrated in mobile phones like the HTC One, 5G Wi-Fi addresses today's data-hungry needs of consumers who increasingly watch video on their mobile devices and/or have multiple wireless devices competing for connectivity," explains Rahul Patel, vice president, wireless connectivity combos (wcc), mobile and wireless group, Broadcom Corporation.

"Wi-Fi (and even Bluetooth) tries to maximize throughput by selecting the most complex coding scheme that the link quality will support. It automatically switches to less complex coding schemes as the link quality declines, compromising data rate for maintaining link," explains Prem Kumar Arora, product marketing manager, wireless, Cypress Semiconductor.

The best fit. Applications like factory automation, home connectivity and media transfer are easily supported by this technology. It has grown to